

Trainings ▾

General Information

Note : There are two different piston and liner combinations available for this engine platform. The ferrous cast ductile (FCD) piston must be used in combination with a nitrided cylinder liner. The steel piston must be used in combination with a high boron liner. It is important that these parts are used in this combination stated and **never** mixed. An engine must **never** be run with a mixture of FCD and steel pistons. It must contain **only** a full set of FCD pistons or a full set of steel pistons. The liners can be distinguished from each other by appearance. The nitride liners will have a matte appearance on the outer circumference. The high boron liners are bright and shiny in appearance.

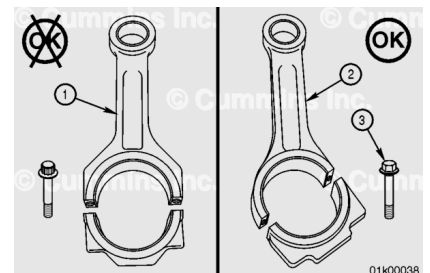
Remove

Use the following steps for engines containing angle split connecting rods (2) **only**.

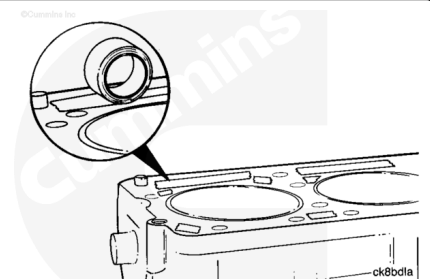
For engines containing straight split connecting rods (1). Refer to Procedure 001-045 in Section 1.

(/qs3/pubsys2/xml/en/procedures/56/56-001-045-tr.html)

Note : The angle split connecting rod (2) can be identified by 6 point head capscrews (3) and the angled joint.



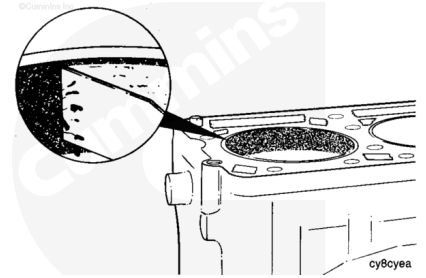
Use tape to protect the push rod galleys, coolant passages, and oil passages from contamination.



⚠ CAUTION ⚠

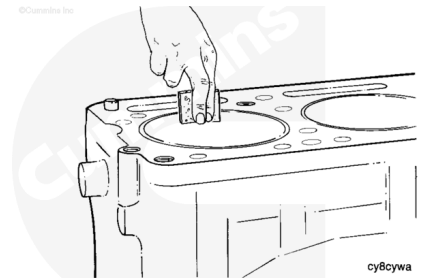
Do not use abrasive paper to remove the carbon deposits. Small particles of abrasive paper will cause severe engine damage.

Use a scraper or a similar blunt-edged tool to loosen the carbon deposits.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's instructions for use. Wear goggles and protective clothing to avoid personal injury.



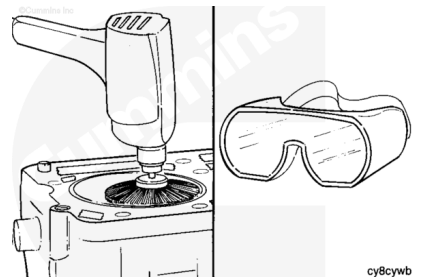
⚠ CAUTION ⚠

Make sure that no abrasive cleaners or materials are used in the piston ring travel area.

Remove the remaining carbon with an abrasive nylon pad, such as Scotch-Brite™ 7448, Part Number 3823258, and solvent. The carbon **must** be removed, but the surface does **not** have to look like new metal.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire brush. Be sure the wire brush is rated for the correct rpm if the brush is motor driven. Flying debris and dirt can cause personal injury.

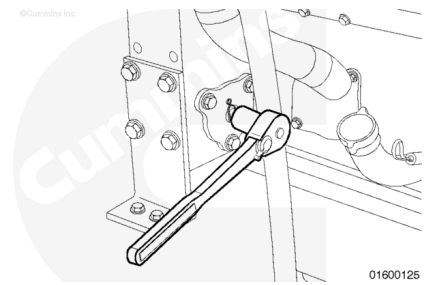


An alternative method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill.

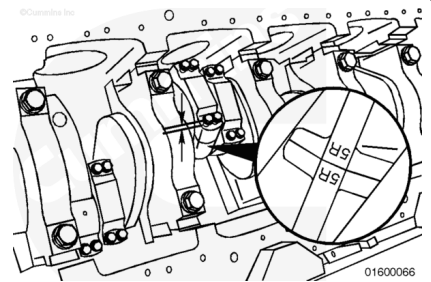
Do **not** use a steel wire wheel of inferior quality. It will lose steel bristles during operation and cause additional contamination.

Do **not** use a steel wire wheel in the piston ring travel area. Operate the wire wheel in a circular motion to remove the deposits.

Use the barring mechanism to rotate the engine. Rotate the crankshaft to position the connecting rod at bottom dead center.



The connecting rods **must** have the cylinder number and bank marked on both the rod and the cap on the side positioned toward the camshaft. Check the rods for correct markings. Use a steel stamp to mark any connecting rod that is **not** correctly marked.

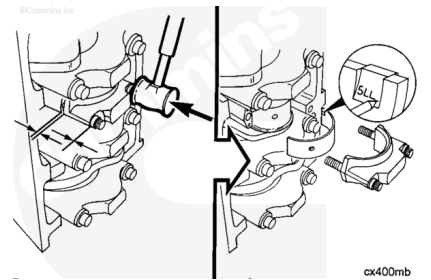


Loosen the capscrews until there is 6 mm [1/4 in] of clearance between the rod cap and the capscrew head.

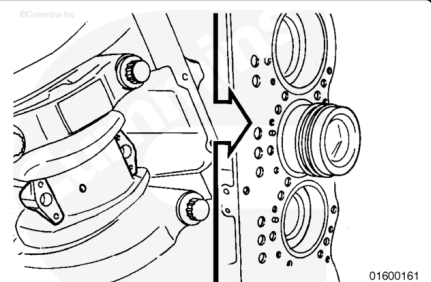
Use a mallet to tap the connecting rod capscrews until the connecting rod cap and rod separate. Remove the capscrews and the cap.

Remove the lower rod bearing. Use an awl to mark the bearing position in the tang area.

Mark the bearing for future identification or analysis.



Push the piston and connecting rod up until the piston rings are above the cylinder liner.



⚠ CAUTION ⚠

Put the piston and rod assembly in a rack to reduce the possibility of damage to the piston and rod assembly.

Remove the piston and connecting rod assembly.

Remove the upper connecting rod bearing.

Use an awl to mark the bearing position in the tang area.

